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ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

NATURAL LANGUAGE PROCESSING [DSA0308]

Campus-Aware Intelligent Lost-and-Found Management System Using NLP

Presented by

A. Syed Thoufiq (192424211)

S. Mohammad Abrar (192472185)

B. Praneeth (192472222)

K. Subrahmanya Venkata Sai Manikanta (192472220)

Guided by:

Dr. T Kumaragurubaran

INTRODUCTION



Campus lost-and-found desks receive a large volume of free-text reports every day.



Manual matching is slow, inconsistent, and easy to miss valid matches.



Different people describe the same object using very different words and phrasing.



NLP techniques can normalize, structure, and compare these descriptions automatically.



The project extracts, analyzes, and ranks matches to reunite items with their owners faster.

LITERATURE REVIEW

S.No	Focus Area	Technique Reviewed	Advantages	Limitations
1	Information Extraction	Named Entity Recognition (NER)	Accurately pulls structured fields from free text	Struggles with informal or very short descriptions
2	Grammatical Analysis	HMM-based POS Tagging	Reliable word-category identification	Accuracy drops on ambiguous or slang wording
3	Feature Representation	TF-IDF + N-gram Extraction	Simple, interpretable, fast to compute	Misses deeper semantic/contextual meaning
4	Similarity Matching	Cosine Similarity	Efficient and widely used for text comparison	Sensitive to vocabulary mismatch between reports
5	Applied Matching Systems	Automated Matching for Classified/Lost-Item Listings	Reduces manual search and turnaround time	Needs reasonably detailed descriptions to work well

PROBLEM STATEMENT

- Lost-and-found reports on campus are currently matched manually by staff, which is slow and error-prone.
- The same item gets described differently by different people, using different words, detail levels, and sentence structures.
- There is a need for an NLP-based system that can understand item descriptions, extract key details, and automatically identify and rank likely matches between lost and found reports.



**Same Item,
Different Words**

OBJECTIVES

1

Extract structured information (item, location, time) from free-text reports.

2

Identify grammatical structure and key word categories using POS tagging.

3

Convert descriptions into numerical features using TF-IDF and n-grams.

4

Compute similarity between lost and found reports using cosine similarity.

5

Rank and recommend the most likely matches for each report.

6

Reduce manual effort and improve turnaround time for reuniting owners with items.

Core NLP Techniques

The pipeline moves from raw text to a ranked match: entities and structure are extracted first, then descriptions are turned into comparable numeric features, and finally scored for similarity.

Example

"iPhone lost near Block C" → item: iPhone, location: Block C → compared to found report → similarity score 0.87 → match



NER



POS Tagging



TF-IDF



Cosine
Similarity

Applications of NLP Techniques in Real World



Spam & text classification



Search engines & information retrieval



Machine translation



Plagiarism detection



Recommendation systems



Chatbots & virtual assistants



Sentiment analysis



Resume / job matching systems



Speech-to-text & voice assistants



Lost-and-found / classified matching (this project)

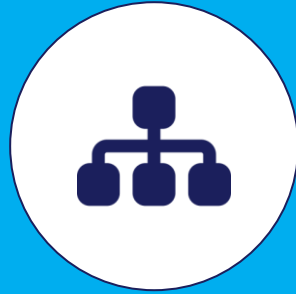
PROJECT MODULES

01



**Campus Context &
Item Information
Extraction**

02



**Syntactic
Structure
Analysis**

03



**Linguistic
Feature
Extraction**

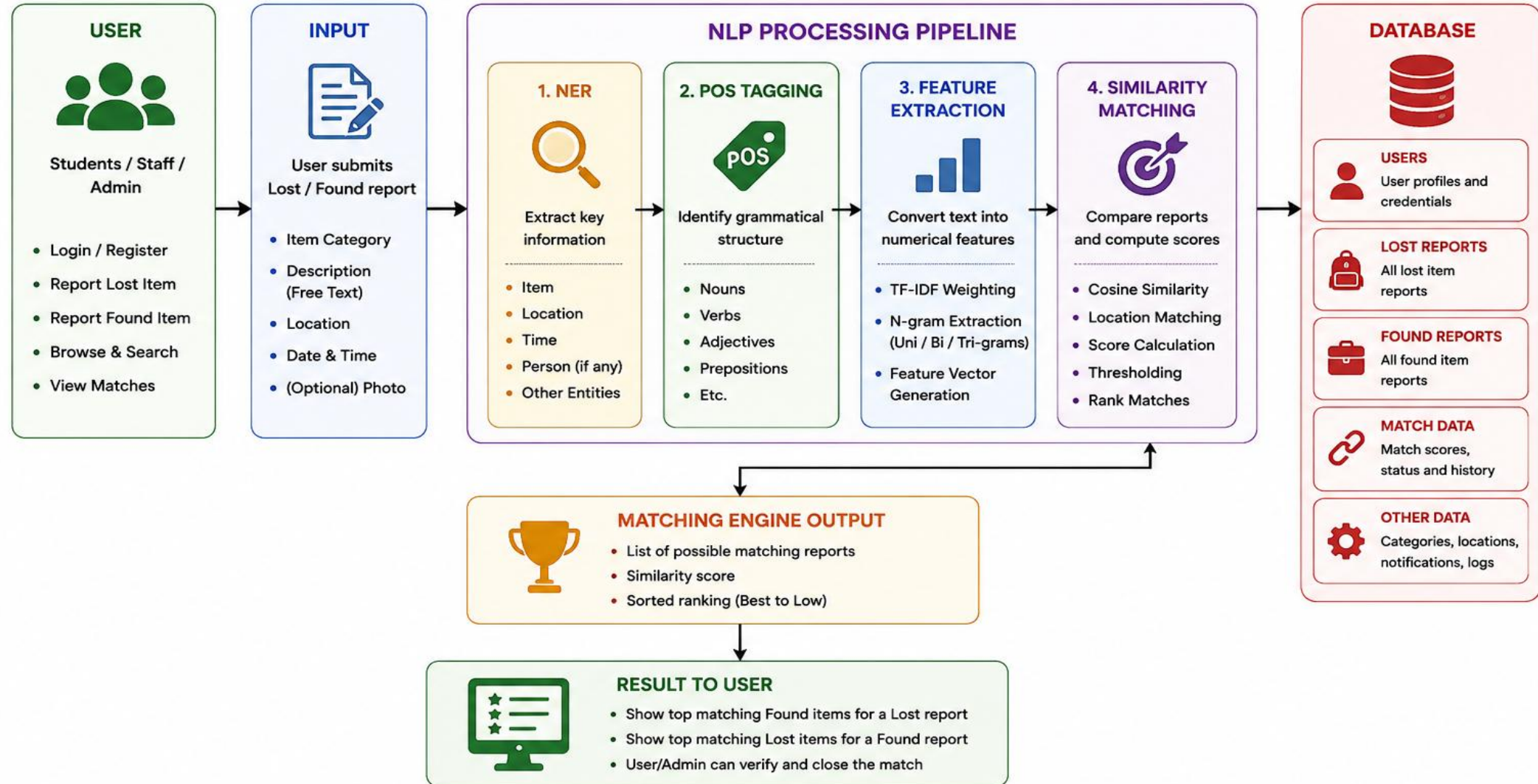
04



**Lost-and-Found
Similarity
Matching**

SYSTEM ARCHITECTURE

Campus-Aware Intelligent Lost-and-Found Management System Using NLP



Module 1

Campus Context & Item Information Extraction

Objective: To extract important information such as item, location, and time from user-submitted lost/found descriptions.

Project Components

- Item Identification
- Location Extraction
- Date / Time Extraction

Tasks

- Parse item name & description
- Detect campus block, room, floor
- Extract reported date/time

Techniques

- Named Entity Recognition (NER)
- Python
- NLP libraries, database



💡 Faculty Viva / Fast Demo Presets (Click to autofill):

+ Blue Wildcraft Backpack (CSE)

+ Sony WH-1000XM4 Headphones (Library)

+ Student ID Card (Admin Block)

1. REPORTER DETAILS

Your Full Name *

Aarav Sharma

Contact Email or Phone *

aarav.s@campus.edu

2. ITEM SPECIFICATIONS & DESCRIPTION

The backend NLP parser will automatically extract named entities (brands, colors) and POS tokens from your description.

Item Category *

Bags & Backpacks

Date Lost

08/28/2026

Free-text Description *

0 chars (NLP vector input)

e.g. "I lost my blue Wildcraft backpack near the CSE block canteen yesterday evening around 5:30 PM. It has my ML notes inside."

Tip: Include specific attributes like color, brand, stickers, or distinguishing marks to maximize cosine similarity matches.

3. CAMPUS LOCATION / LANDMARK

Campus Landmark

CSE Block Canteen

4. PHOTO (OPTIONAL)

Stored client-side as base64



Drag and drop item image here, or browse file

PNG, JPG, WebP up to 5MB

Submitting triggers automatic real-time vector matching against opposite reports.

Submit Lost Item Report →

Module 2

Syntactic Structure Analysis

Objective: To analyze the grammatical structure of the reported description and identify important word categories.

Project Components

- POS Tagging
- Word Category Detection
- Grammar Structuring

Tasks

- Tag nouns, verbs, adjectives
- Identify prepositions & modifiers
- Structure tagged output

Techniques

- HMM-based POS Tagging
- Python
- NLTK



FOUND PROPERTY REGISTRY

Report a Found Item

Help reunite items with their rightful campus owners. Your report will be cross-referenced against all active lost listings.

 Faculty Viva / Fast Demo Presets (Click to autofill):

+ Found Navy Wildcraft Bag (Canteen)

+ Found Sony Wireless Headphones (Library)

+ Found Student ID Card (Admin)

1. REPORTER DETAILS

Your Full Name *

 Aarav Sharma

Contact Email or Phone *

 aarav.s@campus.edu

2. ITEM SPECIFICATIONS & DESCRIPTION

The backend NLP parser will automatically extract named entities (brands, colors) and POS tokens from your description.

Item Category *

 Bags & Backpacks

Date Found

 08/28/2026

Free-text Description *

0 chars (NLP vector input)

e.g. "Found a navy blue Wildcraft laptop bag left unattended at CSE canteen table near the juice stall."

Tip: Include specific attributes like color, brand, stickers, or distinguishing marks to maximize cosine similarity matches.

3. CAMPUS LOCATION / LANDMARK

Campus Landmark

 CSE Block Canteen

4. PHOTO (OPTIONAL)

Stored client-side as base64



Drag and drop item image here, or browse file

PNG, JPG, WebP up to 5MB

Submitting triggers automatic real-time vector matching against opposite reports.

Submit Found Item
Report

Module 3

Linguistic Feature Extraction

Objective: To convert item descriptions into meaningful numerical/textual features for comparison.

Project Components

- TF-IDF Weighting
- N-gram Extraction
- Feature Vector Building

Tasks

- Compute TF-IDF weights
- Generate unigram/bigram/trigram features
- Build comparable vectors

Techniques

- TF-IDF + N-gram Extraction
- Python
- Scikit-learn, NLTK

CAMPUS-WIDE DIRECTORY

Browse Lost & Found Items

Search, filter, and inspect open lost reports and found items across the entire university campus.

 Report Lost

 Report Found

 Search by keywords, brand (Sony, Wildcraft, Dell), roll number, or landmark...

 Grid  Table

REPORT TYPE

All Types (Lost & Found)

CATEGORY

All Categories

CAMPUS LOCATION

All Locations

STATUS

All Statuses

LOST

Bags

Open / Unmatched

I lost my blue Wildcraft backpack near the CSE block canteen yesterday evening around 5:30...

 CSE Block Canteen

 2026-08-25

By Aarav

#st-101

Find Matches →

FOUND

Bags

Open / Unmatched

Found a navy blue Wildcraft laptop bag left unattended at CSE canteen table near the juice...

 CSE Block Canteen

 2026-08-25

By Priya

#nd-201

Find Matches →

LOST

Electronics

Open / Unmatched

Lost my black Sony WH-1000XM4 noise cancelling headphones in Central Library 2nd...

 Central Library (2nd Floor Reading Hall)

 2026-08-26

By Rohan

#st-102

Find Matches →

FOUND

Electronics

Open / Unmatched

Handed over: Sony black wireless over-ear headphones recovered from 2nd floor library...

 Central Library (2nd Floor Reading Hall)

 2026-08-26

By Vikram

#nd-202

Find Matches →

LOST

Id Cards

Matched

Missing student ID card (Roll No. CS-2023-049) with blue campus lanyard near Admin Building...

 Admin Building (Main Entrance)

 2026-08-26

By Ananya

#st-103

Find Matches →

FOUND

Id Cards

Matched

Student ID card found on the stairs of Admin block with blue lanyard. Belongs to CS Dept...

 Admin Building (Main Entrance)

 2026-08-26

By Security

#nd-203

Find Matches →

Module 4

Lost-and-Found Similarity Matching

Objective: To compare lost and found reports and rank the most likely matching items.

Project Components

- Text Similarity Scoring
- Location Matching
- Match Ranking

Tasks

- Compare extracted features
- Compute cosine similarity
- Rank & threshold matches

Techniques

- Cosine Similarity
- Python
- Scikit-learn, NumPy, database



TOTAL LOST REPORTS

5



Logged by students & faculty

TOTAL FOUND ITEMS

5



Deposited at campus desks

RESOLVED THIS WEEK

18



+18% vs last week

ACTIVE NLP PAIRS

10



Similarity > 0.70

Search reports by keyword, user, or spot...

Filter Status: All Open Matched Resolved

TYPE & ID	ITEM DESCRIPTION	CATEGORY	CAMPUS LOCATION	REPORTER / DATE	STATUS	ACTIONS
LOST #st-101	I lost my blue Wildcraft backpack near the CSE block canteen yesterday ...	Bags	CSE Block Canteen	Aarav Sharma 2026-08-25	Open	Matches → Open
FOUND #nd-201	Found a navy blue Wildcraft laptop bag left unattended at CSE canteen table...	Bags	CSE Block Canteen	Priya Patel 2026-08-25	Open	Matches → Open
LOST #st-102	Lost my black Sony WH-1000XM4 noise cancelling headphones in Central Librar...	Electronics	Central Library (2nd Flo...	Rohan Mehta 2026-08-26	Open	Matches → Open
FOUND #nd-202	Handed over: Sony black wireless over-ear headphones recovered from 2nd...	Electronics	Central Library (2nd Flo...	Vikram Singh (Librarian) 2026-08-26	Open	Matches → Open
LOST #st-103	Missing student ID card (Roll No. CS-2023-049) with blue campus lanyard...	Id Cards	Admin Building (Main E...	Ananya Verma 2026-08-26	Matched	Matches → Matched
FOUND #nd-203	Student ID card found on the stairs of Admin block with blue lanyard. Belongs...	Id Cards	Admin Building (Main E...	Security Guard Ramu 2026-08-26	Matched	Matches → Matched
LOST #st-104	Left 65W Dell USB-C laptop charger with power cord in Science Block Lab 4 duri...	Electronics	Science Block Lab 4	Tanmay Das 2026-08-24	Resolved	Matches → Resolved

FUTURE SCOPE

- Develop a mobile application for easy reporting and tracking.
- Improve matching using advanced AI techniques.
- Add real-time notifications for potential matches.
- Integrate QR codes for quick lost-and-found reporting.
- Add campus map and location-based features.
- Support multiple languages for easier use.



**Faster Reunions,
Less Waste**

CONCLUSION

- Uses NLP (NER, POS tagging, TF-IDF, cosine similarity) to automatically match lost and found campus reports.
- Reduces manual effort and speeds up reuniting items with their owners.
- Handles varied user descriptions through structured extraction and similarity ranking.
- Supports smarter, more responsive campus lost-and-found services.
- Contributes to SDG 12: Responsible Consumption and Production by encouraging reuse over replacement.



Faster Reunions,

Less Waste

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- [2] Manning, C. D., Raghavan, P., and Schütze, H. Introduction to Information Retrieval. Cambridge University Press.
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Thank You